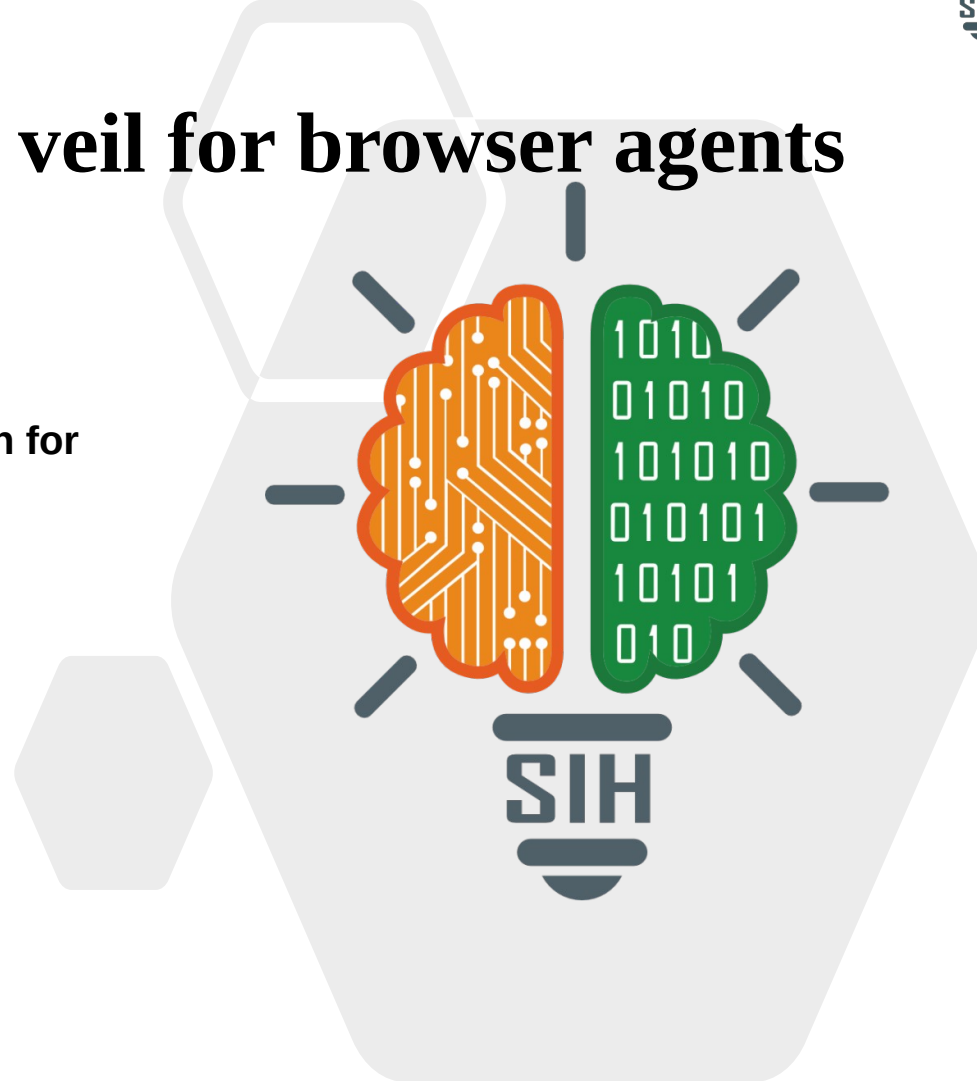


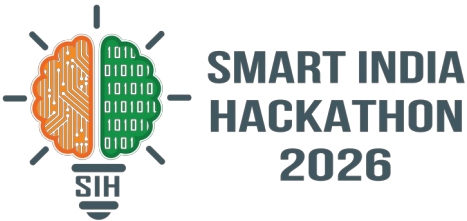
AAVARAN — a privacy veil for browser agents

- Problem Statement ID – 26171
- Problem Statement Title – On-device Visual Perception for Light-weight Browser Agents
- Theme – Smart Automation
- PS Category- Software
- Team ID – <Team ID>
- Team Name (Registered on portal) – <Team name>



<Team
name>

AAVARAN — ON-DEVICE PRIVACY VEIL



Proposed solution

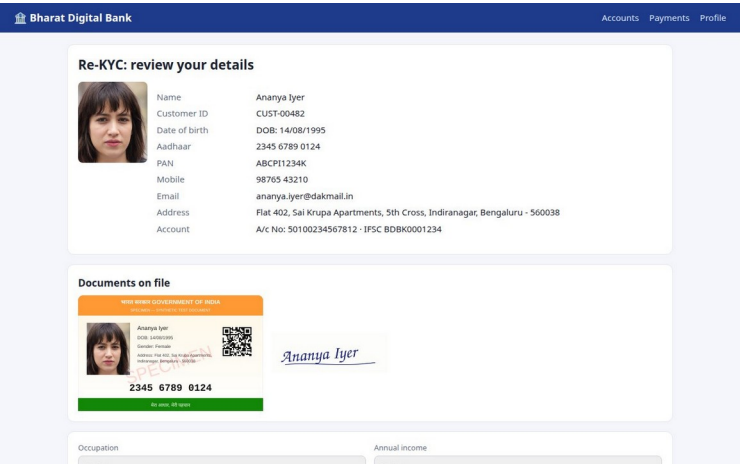
- **Browser extension (Chrome, Edge, Brave, Firefox) + FastAPI server.** Three small models run in the browser (WebGPU → WASM) and read the screen: YuNet faces, MobileCLIP-SO vision transformer, BERT-small PII NER (52 MB total).
- Before any request, **personal data is replaced by consistent tokens** ([NAME_1], [AADHAAR_1], [OTP_1]) and black-boxed at pixel level in the screenshot, with the same labels.
- An open-weights VLM on the server (Qwen2.5-VL / Llama-4 / Gemma-3) plans actions over tokens + numbered UI marks; the client swaps real values back in **on the device**.

How it addresses the problem

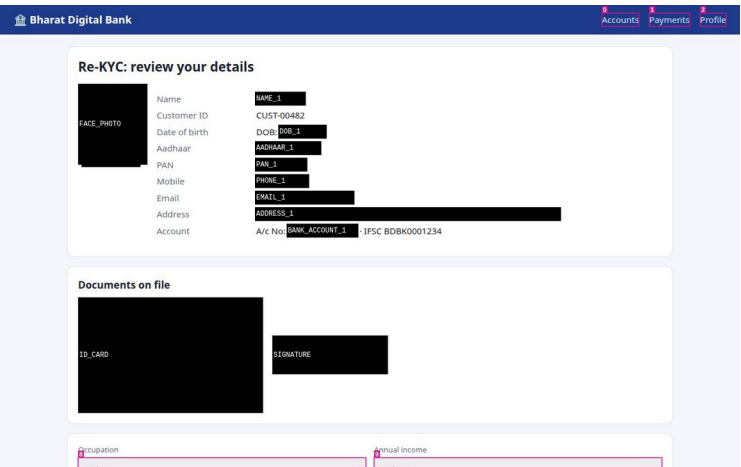
- Server-side agents can't be trusted with Aadhaar cards, OTPs, faces and bank pages; a local-only agent is too weak. Aavaran splits the work: perception + privacy local, reasoning remote.
- Redaction is **verifiable**: a Privacy X-ray shows the exact frame and tokens that leave; the server's own audit log is checked after every task — **0 leaks** in 5 end-to-end tasks.

Innovation & uniqueness

- **DOM-grounded pixel redaction:** PII found in text is mapped to exact screen rectangles (Range.getClientRects) — no OCR guessing; vision covers pixels the DOM can't describe (faces, ID-card photos, signatures).
- **Reversible pseudonym vault:** the agent can fill your form with your details without ever seeing them.
- **Pixels × structure:** a ViT fused with DOM structure understands unseen websites 73.3% vs 55.8% from pixels alone.
- **India-first PII:** Aadhaar (Verhoeff), PAN, GSTIN (mod-36), UPI, IFSC-context accounts, OTPs.

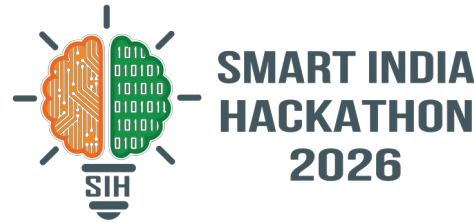


What the user sees

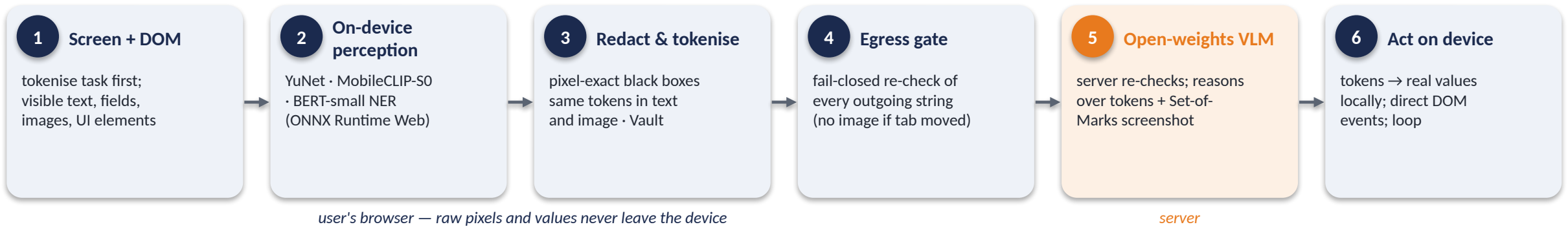


What the server receives — redacted on the device

TECHNICAL APPROACH



<Team
name>



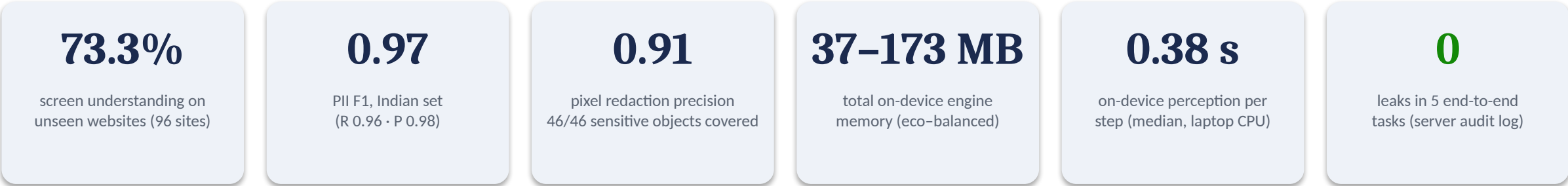
Technologies

Client	Manifest V3 extension, JavaScript, esbuild; Chrome/Edge/Brave + Firefox from one source
On-device ML	ONNX Runtime Web — WebGPU, multi-threaded WASM SIMD (cross-origin isolated); offscreen document hosts all models once
Models	YuNet FP32 0.2 MB · MobileCLIP-S0 image tower FP16 23 MB (prompt embeddings precomputed) · BERT-small PII INT8 29 MB
Server	Python FastAPI; OpenAI-compatible open-weights VLM (vLLM / Ollama): Qwen2.5-VL-7B, Llama-4-Scout, Gemma-3
Evaluation	Puppeteer + real Chrome, onnxruntime-node; WIDER FACE, ai4privacy, 217-screen web set

Methodology (built and measured)

- **Text PII → pixels:** visible text grouped per block, scanned by checksum rules + NER + known Vault values; each span → Range.getClientRects() → black box labelled with its token.
- **Vision:** all on-page images packed into one 640² YuNet mosaic (avatars become detectable); one batched MobileCLIP pass for screen state + image regions (ID card, signature, QR, face photo), fused with DOM cues.
- **Adaptive compute:** dHash frame cache, per-page screen cache, NER/region caches, eco / balanced / max modes, models warmed while the user types.
- **Grounding:** numbered Set-of-Marks on interactive elements; actions reference node ids; tokens rehydrated only at execution.
- **Defence in depth:** sensitive-field values never serialised; egress gate; server-side re-check (Aadhaar/Verhoeff, PAN, UPI, Luhn); audit log.

Working prototype, measured against all five evaluation criteria (reproducible scripts in eval/):



Challenges, risks and how we handle them

Challenge	Risk	Strategy (implemented)
Low-end client devices	Models too heavy / slow	52 MB total; measured precision choice (FP32 YuNet 2.6× faster than INT8 in WASM); eco mode 37 MB; caches make an unchanged frame ~0.15 s; WebGPU when present
Missed PII = privacy leak	Rules or NER miss a value	Four layers: checksum rules + NER + Vault-guided matching + sensitive-field boxing; fail-closed egress gate; server re-check; audit log
Over-redaction	Agent loses context	Typed, consistent tokens keep structure; checksum validation keeps order ids, PNRs, prices, IFSC readable (1/120 false alarms on hard negatives)
Unfamiliar UIs	ViT not trained on screens	Fuse pixels with DOM structure (+17.5 pts on unseen sites); element grounding via DOM + Set-of-Marks is exact
Server model availability	Cloud dependence	Any OpenAI-compatible open-weights endpoint (vLLM / Ollama offline); stateless server; mock mode for demos

Who benefits

- 1 Citizens** — use AI assistants on DigiLocker, UPI, bank, e-KYC and government portals without handing Aadhaar, OTPs or faces to a server
- 2 Government & ISRO / DoS** — automate internal web workflows on sensitive screens; the server can run on-premise with open weights
- 3 Enterprises & startups** — build agents that comply with the DPDP Act 2023 by design (data minimisation, purpose limitation, auditability)
- 4 Low-end & rural devices** — runs on a 4-core laptop CPU with no GPU; small payloads (~50 KB image per step) suit slow networks

Benefits

- **Social:** trust in AI assistance; privacy that users can inspect (Privacy X-ray).
- **Economic:** open-weights models, no per-token vendor lock-in; smaller prompts (tokens, one ~50 KB JPEG).
- **Strategic:** sovereign, offline-deployable stack, aligned with Atmanirbhar Bharat and DPDP.
- **Environmental:** caches skip repeated inference; less data moved per step.

Screen understanding, unseen websites



PII recall, Indian set (precision ≥ 0.98)



Redaction on demo sites (pixels)



Research & models

- Vasu et al., MobileCLIP: Fast Image-Text Models through Multi-Modal Reinforced Training, CVPR 2024 — arxiv.org/abs/2311.17049
- Wu et al., YuNet: A Tiny Millisecond-level Face Detector, Machine Intelligence Research 2023 — github.com/opencv/opencv_zoo
- Yang et al., Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V, 2023 — arxiv.org/abs/2310.11441
- Bai et al., Qwen2.5-VL Technical Report, 2025 — arxiv.org/abs/2502.13923 (open-weights server VLM)
- Radford et al., Learning Transferable Visual Models from Natural Language Supervision (CLIP), ICML 2021
- ONNX Runtime Web (WebGPU / WASM) — onnxruntime.ai · [Transformers.js](https://huggingface.co/docs/transformers.js) — huggingface.co/docs/transformers.js
- BERT-small PII detection (ONNX) — huggingface.co/onnx-community/bert-small-pii-detection-ONNX
- Datasets: WIDER FACE (Yang et al., CVPR 2016); [ai4privacy/pii-masking-200k](https://ai4privacy.org/pii-masking-200k)
- Digital Personal Data Protection Act, 2023 (MeitY); UIDAI Aadhaar number checksum (Verhoeff)

Prototype & evidence

- **Code:** github.com/Malaybhai11/SIH2026-SIH26171
- **Extension** for Chrome/Edge/Brave (dist/) and Firefox (dist-firefox/)
- **Demo sites** (synthetic data, AI-generated faces): bank KYC, webmail OTP, social feed, checkout, ISRO outreach registration
- **Scorecard:** [eval/results/SUMMARY.md](#) — every number reproducible with one npm script
- **End-to-end tasks:** checkout 7 s, inbox 6 s, kyc 22 s, register 17 s, social 6 s — all passed, 0 leaks
- **Faces (WIDER val):** P 0.78 / R 0.73; ai4privacy PII F1 0.85